

PoolMate

Detailed User Guide — equimolar library pooling, dilution planning, and Eppendorf epMotion worklist generation. Everything runs locally in your browser; no data leaves your machine.

Overview

PoolMate plans equimolar pools and dilutions and generates the epMotion worklist CSVs (DNA.csv / Buffer.csv). Enter your samples once, then pick the depth of output you need — from a quick molarity check to a full robot worklist.

The three workflows

- **Calculate Molarities** — uses only the Sample section to convert concentration + fragment size into nM. No pooling volumes are produced. Minimum input: concentration + size, or direct nM.
- **Quick Pool** — a manual pooling table from total fmol or fmol per sample. No presets, racks, buffer positions, or worklists required. Best for fast bench calculations and hand pipetting.
- **Generate Worklists** — the full epMotion workflow: sample molarities, pool target, layout, deck settings, buffer, and tips, exporting DNA.csv and Buffer.csv for robot execution.

Sample section vs. create-pool parameters

The **Sample section** describes what you currently have: names, source wells, concentrations, fragment sizes, and any direct nM values. The **create-pool parameters** describe the pool you want: target nM or fmol, final volume, and layout (plate vs. single tube).

Input modes

- **Mass mode (ng/μL + bp)**: enter a concentration and fragment size; PoolMate computes molarity. Replicate Qubit reads can be entered separated by commas, semicolons, or spaces, and are averaged automatically.
- **Molar mode (direct nM)**: enter molarity directly when it has already been measured.
- The **Mass-mode samples** counter in the summary shows how many rows are in mass mode with a concentration entered; switch a row's Mode to direct nM to drop it from that count.

Standalone dilution calculator

Dilute one sample or a whole list to a target concentration (nM or ng/μL), completely independent of pooling. Enter each sample's stock concentration; every other parameter defaults to the shared values at the top of the section but can be overridden per row:

- **Target and final volume** — the concentration and volume to dilute to.
- **Min transfer** — the smallest stock volume you can reliably pipette.
- **Max well/tube capacity** — the largest volume a destination well can hold.
- **Round to** — the increment volumes are rounded to.

Any blank cell in a row falls back to the matching shared default. The calculator shows DNA + diluent volumes per sample, the total diluent for the batch, and a tube graphic per sample with a Done checklist for the bench.

Serial dilution

When a sample is too concentrated to reach the target in one step within the transfer and capacity limits, PoolMate lays out the serial steps automatically (equal split factor, chained wells). For pooling, this is exported as a separate ordered Dilution.csv that runs first — the robot mixes each well — and the pool then draws from the diluted wells.

Plate map and tube preview

The live plate map mirrors the source wells from the sample table, with sample names shown in the selected wells, so you can confirm rack loading before the run. A Plate / Tube toggle also previews a single pooled tube as one translucent, labelled colour band per sample.

Output files

- **DNA.csv** — the DNA transfer worklist for the epMotion.
- **Buffer.csv** — the buffer transfer worklist.
- **Dilution.csv / predilution plan** — produced when pre-dilution is needed; load Dilution.csv first, then DNA.csv, then Buffer.csv.
- **Worksheet** — the full calculation record, including the post-rounding CV check.
- **Inputs** — the saved run settings, re-importable into PoolMate.

About the numbers

Molarity for dsDNA: $\text{nM} = \text{conc}(\text{ng}/\mu\text{L}) \times 10^6 / (660 \times \text{size_bp})$. Transfers are calculated as equal fmol per sample, then rounded to the selected increment and audited for post-rounding deviation (CV%). In fmol mode the tool derives target nM as total fmol divided by final volume.

Important: The built-in platform presets are vendor starting points. Confirm all volumes and loading concentrations against your own kit / instrument SOP before running anything on the robot.